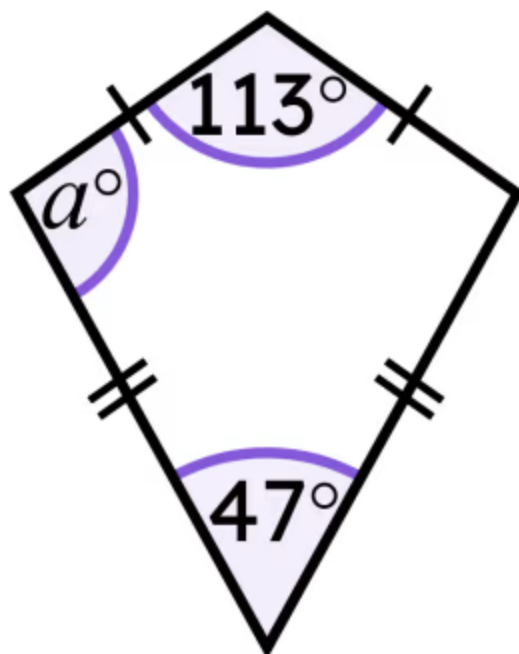




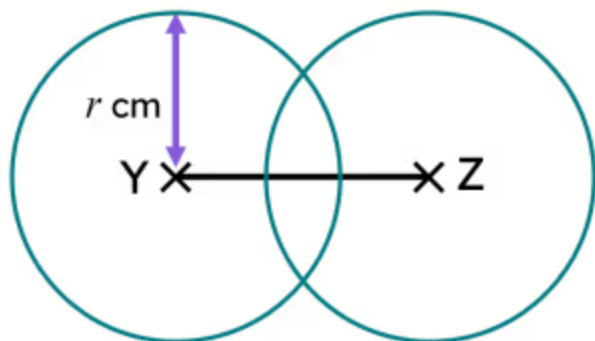
Perpendicular to a given line through a given point

- 1 This quadrilateral is a kite. Find the value of angle a , in degrees. Fill in the blank



100

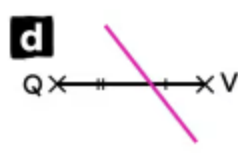
- 2 This diagram shows an accurate construction of a perpendicular bisector. YZ has a length of 8.4 cm. Which of these is the smallest value of r that can be used in this construction?
Tick 1 correct answer



- ☐ 2.1 cm
☒ 4.3 cm
☐ 8.4 cm
☐ 16.8 cm
☐ 8.4π cm



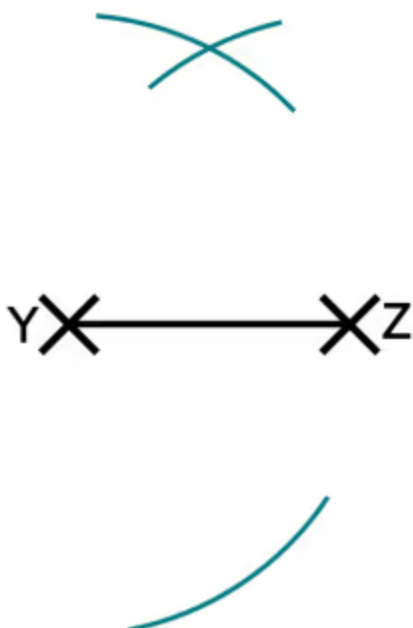
3 Match the sentence to the diagram. Write the correct letter in each box



a	a
b	b
c	c
d	d

<i>b</i>	A perpendicular to line segment QV
<i>a</i>	A bisector to line segment QV
<i>c</i>	A perpendicular bisector to line segment QV
<i>d</i>	An intersecting line to line segment QV

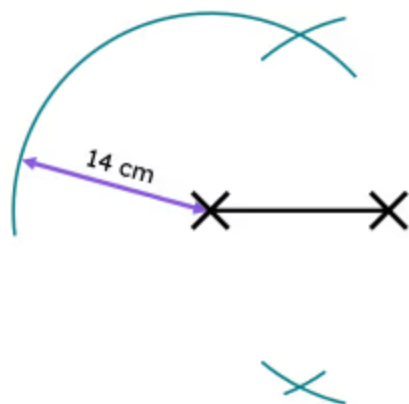
4 This is an incomplete construction to find the perpendicular bisector of YZ. Which of these steps is missing from this construction? Tick **2** correct answers



- ☐ A circle with centre at point Y.
- ☐ A line segment connecting the intersection to point Z.
- ☒ An arc below YZ with centre at point Z.
- ☐ An arc above YZ with centre at point Z.
- ☒ A line through two pairs of intersecting arcs.

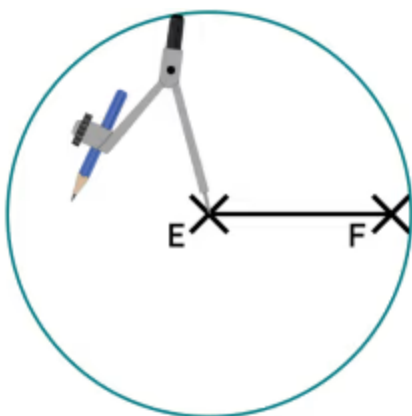


- 5 This is an accurate construction to draw a rhombus. What is the perimeter (in cm) of the rhombus that can be drawn from this construction? Fill in the blank



56

- 6 Sofia starts her construction of a perpendicular bisector to EF. She is about to move the pair of compasses to construct a second circle. Which of these statements are true? Tick 2 correct answers



- ☐ Sofia needs to place the compass needle at the midpoint of EF.
- ☐ Sofia should draw a circle with centre at the intersection of EF and the circle.
- ☐ Sofia needs to draw an arc with centre at point F next.
- ☒ Sofia should adjust the compass width so that both circles drawn are congruent.
- ☒ Currently, the next circle Sofia draws will be smaller than the first.

